

Drone Project - Personal Journal and Team Feedback

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Personal record
Primary Role	Team lead and system integrator
Original Source	drone-personal-journal-and-feedback.pdf (20 pages)

Executive Summary

This personal record reflects the leadership and execution process behind the drone project. It emphasizes weekly progress, team coordination, design tradeoffs, and the pressure of turning a complex build into a working final system.

For graduate review, the value of this document is not only the technical result. It shows that I can keep a team moving when requirements are ambiguous, failure modes overlap, and final evaluation pressure is real.

Applicant Contribution Signals

- Coordinated mechanical, electrical, control, and testing tasks across the team.
- Helped translate weekly setbacks into concrete next actions.
- Balanced ambition with practical risk control near final evaluation.

Leadership Evidence

The project required repeated communication between subteams because airframe changes affected payload behavior, electronics layout, and flight-test safety.

Leadership meant making the system testable: deciding what to try next, what to repair first, and when to simplify.

Reflection

This project strengthened my interest in robotics because it made clear that physical systems reward engineers who can connect design, control, testing, and people.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Ambiguous failures	Organized next tests and subsystem checks	Kept progress from stalling	Project leadership
Limited time	Prioritized final-run reliability	Protected the strongest route to success	Decision making
Team coordination	Connected mechanical and control discussions	Improved shared understanding	Technical communication

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.